

Answers to Questions from T 7.2.1

Name: Tran Quang Huy

Student ID: 106212636

Hand execution using a trace table

Demonstrate how the following code is executed in the computer.

Name of the function: find_the_value

```
def whatShouldThisFunctionBeCalled(data, val):  
    i = 0  
    result = False  
    while i < len(data):  
        if data[i] == val:  
            result = True  
            return result  
        i += 1  
    return result
```

Draw and complete trace tables for the following two sets of data and place the final result in the table below:

<i>data</i>	<i>val</i>	<i>Result</i>
[2, 6, -4, 3, 7]	3	
[-2, 8, 2, -5, 9]	6	

Result 1: True

Result 2: False

Once you have completed your trace tables (below) write the name you chose for the function above:

Place your trace tables below:

For data set: **data** = [2, 6, -4, 3, 7] **val** = 3

Loop	i	Data[i] and val comparison	Result
First loop	0	0	False
1	1	2 != 3	False
2	2	6 != 3	False
3	3	-4 != 3	False
4	4	3 = 3	True

For data set: **data** = [-2, 8, 2, -5, 9] **val** = 6

Loop	i	Data[i] and val comparison	Result
First loop	0	0	False
1	1	-2 != 6	False
2	2	8 != 6	False

3	3	$2 \neq 6$	False
4	4	$-5 \neq 6$	False
5	5	$9 \neq 6$	False